

EXPERIMENT-12 Tic Tac Toe Game

Aim

To implement the Tic Tac Toe game using Python.

Objectives

- To understand game-playing AI concepts.
- To implement a two-player game.
- To determine the winner or draw.

Algorithm

1. Start.
2. Create an empty board.
3. Allow Player X and Player O to play alternately.
4. Check for a winning condition after every move.
5. If a player wins, declare the winner.
6. If all positions are filled, declare a draw.
7. Stop.

Flow Chart

Start

|



Create Board

|

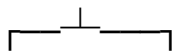


Player Move

|



Winner?



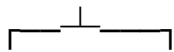
Yes No



Display Winner



Board Full?



Yes No



Draw Next Player



Program

```
board = [' ']*9
```

```
def print_board():
```

```
    print()
```

```
    print(board[0],'|',board[1],'|',board[2])
```

```
    print("---+---+---")
```

```
    print(board[3],'|',board[4],'|',board[5])
```

```
    print("---+---+---")
```

```
    print(board[6],'|',board[7],'|',board[8])
```

```
print()
```

```
def check(player):
```

```
    wins = [  
        [0,1,2],[3,4,5],[6,7,8],  
        [0,3,6],[1,4,7],[2,5,8],  
        [0,4,8],[2,4,6]  
    ]
```

```
    for w in wins:
```

```
        if all(board[i]==player for i in w):
```

```
            return True
```

```
    return False
```

```
player = 'X'
```

```
for turn in range(9):
```

```
    print_board()
```

```
    pos = int(input(f"Player {player}, Enter position (1-9): ")) - 1
```

```
    if board[pos] == ' ':
```

```
        board[pos] = player
```

```
    if check(player):
```

```

print_board()

print(player,"Wins!")

break

```

```

player = 'O' if player == 'X' else 'X'

```

```

else:

```

```

    print("Position already occupied.")

```

```

else:

```

```

    print_board()

```

```

    print("Match Draw")

```

Output

```

main.py
1  board = [' ']*9
2
3  def print_board():
4      print()
5      print(board[0],'|',board[1],'|',board[2],'|',board[3],'|',board[4],'|',board[5])
6      print("-----")
7      print(board[3],'|',board[4],'|',board[5],'|',board[6],'|',board[7],'|',board[8])
8      print("-----")
9      print(board[6],'|',board[7],'|',board[8])
10     print()
11
12     def check(player):
13         wins = [
14             [0,1,2],[3,4,5],[6,7,8],
15             [0,3,6],[1,4,7],[2,5,8],
16             [0,4,8],[2,4,6]
17         ]
18
19         for w in wins:
20             if all(board[i]==player for i in w):
21                 return True
22         return False
23
24     player = 'X'
25     for turn in range(9):
26         print_board()
27         pos = int(input(f"Player {player}, Enter position (1-9): "))
28         if board[pos] == ' ':

```

```

| |
+---+
| |
+---+
| |
Player X, Enter position (1-9): 5
| |
+---+
| X |
+---+
| |
Player O, Enter position (1-9):

```

Result

The Tic Tac Toe game executed successfully and correctly identified the winner or draw.

Conclusion

The program successfully simulated a two-player Tic Tac Toe game by validating moves and determining the game's outcome.